Python (part2)

Univariate Plots

```
dataset.hist()
```

Multivariate Plots

```
from pandas.plotting import scatter_matrix

scatter_matrix(dataset)
```

Generating 2D classification dataset

```
from sklearn.datasets.samples_generator import make_blobs

X, Y = make_blobs(n_samples=100, centers=2, n_features=3, random_state=1)

for i in range(len(X)):

        print("X=%s, Y=%s" % (X[i], Y[i]))
```

Generating 2D regression dataset

```
from sklearn.datasets import make_regression

X, Y = make_regression(n_samples=4, n_features=2, noise=0.1)

for i in range(len(X)):

        print("X=%s, Y=%s" % (X[i], Y[i]))
```

If you have any query on a specific method, have a look on the online documentation. For example:

https://scikit-learn.org/stable/modules/generated/sklearn.datasets.make_regression.html

More on prediction

1. Class Predictions in classification

```
from sklearn.linear_model import LogisticRegression
from sklearn.datasets.samples_generator import make_blobs
X, Y = make_blobs(n_samples=100, centers=2, n_features=2, random_state=1)
model = LogisticRegression()
model.fit(X,Y)
Xnew = [[-0.79415228, 2.5]]
Ynew = model.predict(Xnew)
print("X=%s, Predicted=%s" % (Xnew[0], Ynew[0]))
Xnew2, _ = make_blobs(n_samples=3, centers=2, n_features=2, random_state=1)
Ynew2 = model.predict(Xnew2)
# show the inputs and predicted outputs
for i in range(len(Xnew2)):
        print("X=%s, Predicted=%s" % (Xnew2[i], Ynew2[i]))
```

2. Probability Predictions in classification

This function is only available on those classification models capable of making a probability prediction, which is most, but not all, models.

```
from sklearn.linear_model import LogisticRegression
from sklearn.datasets.samples_generator import make_blobs
# generate 2d classification dataset
X, y = make_blobs(n_samples=100, centers=2, n_features=2, random_state=1)
# fit final model
model = LogisticRegression()
model.fit(X, y)
# new instances where we do not know the answer
Xnew, _ = make_blobs(n_samples=3, centers=2, n_features=2, random_state=1)
# make a prediction
ynew = model.predict_proba(Xnew)
# show the inputs and predicted probabilities
for i in range(len(Xnew)):
        print("X=%s, Predicted=%s" % (Xnew[i], ynew[i]))
```

3. Predictions with regression models

```
from sklearn.linear_model import LinearRegression
from sklearn.datasets import make_regression
# generate regression dataset
X, Y = make_regression(n_samples=10, n_features=2, noise=0.1)
# fit final model
model = LinearRegression()
model.fit(X, Y)
# new instances where we do not know the answer
Xnew, _ = make_regression(n_samples=2, n_features=2, noise=0.1, random_state=1)
# make a prediction
Ynew = model.predict(Xnew)
# show the inputs and predicted outputs
for i in range(len(Xnew)):
        print("X=%s, Predicted=%s" % (Xnew[i], Ynew[i]))
```